

RustyClean: a Rust pipeline for adaptive minimizer depletion and conditional alignment verification in short-read metagenomics

APPLICATION NOTE

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Abstract

Summary: Host-read depletion must retain microbial signal while removing abundant host DNA. RustyClean is a Rust workflow that combines fastp quality control, Deacon minimizer-based depletion and conditional Bowtie2 verification. Verification is triggered only when Deacon reports a removed fraction above a configurable threshold. On six simulated metagenomes spanning 1–90% host, RustyClean AUTO was 3.45–10.19-fold faster than KneadData and 1.29–2.30-fold faster than fastp+Hostile; with verification disabled, it remained 1.05–2.59-fold faster than fastp+Hostile. With verification, AUTO achieved $F1 \geq 0.998$, 0.0000% host carry-over and $\leq 0.31\%$ microbial loss on high-host libraries.

Availability and implementation: RustyClean is implemented in Rust, uses fastp, Deacon, Bowtie2 and samtools, and is available under the MIT licence at <https://github.com/HuangShiLab/rustyclean>. Benchmark source data are available at <https://github.com/HuangShiLab/rustyclean-paper>.

Supplementary information: Full benchmark tables and figure-generation code are available in the project repository. A detailed benchmark manuscript is available as a separate bioRxiv-ready submission.

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1 Introduction

Short-read metagenomes from host-associated sites often contain abundant host DNA. Alignment-based depletion is sensitive but can remove genuine microbial reads, whereas rapid classifier- or minimizer-based filters are faster but may retain host reads (Gao et al., 2025). Deacon is a minimizer-based sequence filter with near-alignment accuracy and host-fraction-independent speed (Constantinides et al., 2025). However, it is a depletion primitive rather than a complete cohort workflow. RustyClean addresses this gap by adding quality control, adaptive verification, checkpointing, memory-aware concurrency and output validation.

2 Features and implementation

RustyClean AUTO processes each sample in four ordered stages. fastp first performs paired-end adapter detection and quality trimming (Chen et al., 2018). Deacon v0.17.0 then depletes host reads against a compact pangenome index. RustyClean parses the Deacon summary and passes the retained reads to Bowtie2 only when the reported removed fraction is at least 0.30; below this threshold, the Deacon output is accepted without a second alignment pass (Langmead and Salzberg, 2012). Finally, automated checks reject truncated outputs or samples exceeding a configurable residual-host limit.

The workflow is implemented as a single Rust binary and provides sample manifests, atomic checkpoints, content-invalidating input fingerprints, bounded sample-level concurrency and memory-aware worker capping. Each worker maps the same read-only Deacon index, so additional workers add little index memory. In a 16-sample cohort, increasing concurrency from one to eight workers reduced wall time from 2364 s to 302 s, a 7.84× speedup at 98% parallel efficiency, while the median resident set size per Deacon worker remained approximately 4.52 GiB. The streamlined interface intentionally exposes the production AUTO pipeline and a Deacon-only benchmarking mode rather than multiple deprecated depletion backends.

3 Results

On six simulated single-end metagenomes generated with InSilicoSeq and scored with per-read ground truth, RustyClean AUTO required 90.4–2396.7 s across 5–100 million reads (Gourlé et al., 2019). This was 3.45–10.19-fold faster than KneadData v0.12.3 and 1.29–2.30-fold faster than fastp+Hostile (Constantinides et al., 2023; McIver et al.,

2018). With Bowtie2 verification disabled, the complete fastp+Deacon workflow averaged 141.9–1483.7 s and remained 1.05–2.59-fold faster than fastp+Hostile across all six datasets. Deacon depletion alone took 9.6–202.4 s and was 40.9–130.2-fold faster than the complete KneadData pipeline. The Hostile comparator is the sequential fastp+Hostile workflow: the plotted times are the sums of independently timed fastp and Hostile stages, not Hostile alone. Across the panel, the Hostile stage alone required 6.17–20.30-fold longer than Deacon depletion only. Peak memory was 4.54 GiB and was bounded by the pangenome index rather than sample size. Cohort-level concurrency scaled near-linearly because all workers shared the same memory-mapped index (Fig. 1c–e).

On four high-host datasets (50–90%), Bowtie2 verification reduced host carry-over to 0.0000% of retained output, whereas fastp+Hostile retained 0.17–0.68% and KneadData retained 0.25% (Table 1). RustyClean AUTO maintained F1 ≥ 0.998 and lost at most 0.31% of microbial reads. Verification therefore provides a practical route to zero measured host carry-over without applying an alignment pass to every low-host sample. If users prefer the faster fastp+Deacon configuration, disabling recheck preserved F1 ≥ 0.99881 and microbial loss $\leq 0.238\%$ across the panel, but retained the Deacon host residue (0.0035–0.0037% on high-host libraries).

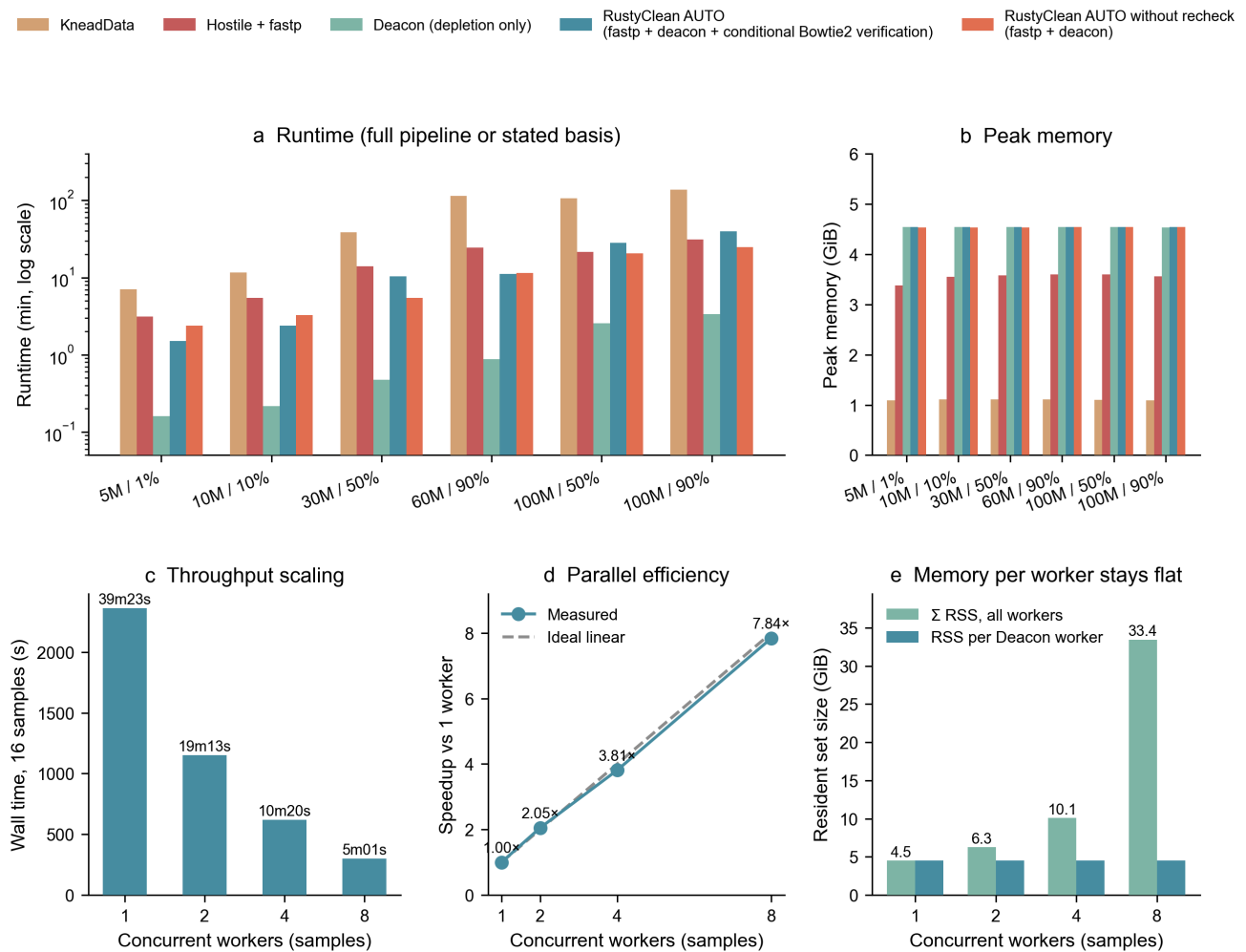


Fig. 1. Performance and parallel scaling of RustyClean. (a,b) Runtime and peak memory for five depletion workflows or arms on six simulated metagenomes; runtime uses a logarithmic scale. KneadData, fastp+Hostile and both RustyClean AUTO arms are full pipelines, whereas Deacon is depletion only. The Hostile+fastp bar is the sum of separately timed fastp and Hostile stages. (c–e) Sixteen 10M-read samples processed with one to eight concurrent workers: cohort wall time, speedup relative to one worker, and resident set size.

Table 1. Accuracy on four high-host simulated datasets.

Workflow	F1 range	Microbial loss (%)	Host carry-over (%)
KneadData	0.97603–0.99098	1.425–2.154	0.2493–0.2500
fastp+Hostile	0.96522–0.99878	0.000–0.039	0.1734–0.6758
Deacon only	0.99920–0.99997	0.000–0.121	0.0035–0.0037
RustyClean AUTO	0.99845–1.00000	0.000–0.310	0.0000

Workflow	F1 range	Microbial loss (%)	Host carry-over (%)
RustyClean AUTO without recheck	0.99920–0.99997	0.0055–0.1211	0.0035–0.0037

4 Conclusion

RustyClean turns rapid minimizer depletion into a reproducible host-depletion workflow. Conditional Bowtie2 verification preserves the speed advantage of Deacon while removing residual host signal precisely in the high-host samples where it matters most.

Availability and implementation

RustyClean is available under the MIT licence at <https://github.com/HuangShiLab/rustyclean>. The benchmarked research source is archived at commit [7ab1a4b](#); the streamlined AUTO interface is at commit [55f93af](#). Figure source data and analysis code are available at <https://github.com/HuangShiLab/rustyclean-paper>.

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